

Implementation of FIR Filters Using Rectangular Window

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Filters are an important part of electronic circuits today. Whether it is the analogue or digital world, the role of filters is very important. Presented here is a C++ program for an FIR filter design using the rectangular window technique.

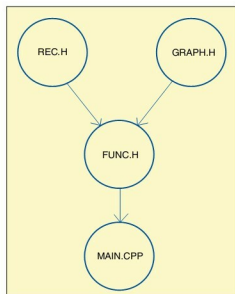


Fig. 1: Software structure

However, before going deep into the designing of software for filters, it is necessary to understand the basic concept of filters. In both the analogue and digital worlds, the signal comes coupled with noise interruptions. To get rid of this unwanted noise, we use filters. Note that sometimes filters not only remove the unwanted signals but also some parts of the signal itself. For example, our speech signals have a bandwidth of 20 kHz but these are filtered up to 3.4 kHz without the loss of intelligence.

There are different types of filters available in the market:

Low-pass filter. This filter passes all the frequencies lower than a defined frequency (say f_1 Hz) and rejects



all other frequencies. Since it passes all the frequencies lower than f_1 Hz, it is called a low-pass filter.

High-pass filter. This filter passes all the frequencies above f_1 Hz and rejects all other frequencies. Since it passes all the frequencies above f_1 Hz, it is called a high-pass filter.

Band-pass filter. This filter passes all the frequencies in between two frequencies, say, f_1 and f_2 and rejects all other frequencies. Since it passes a band of frequencies, it is called a band-pass filter.

Band-reject filter. This filter passes all the frequencies, except the ones falling between two frequencies f_1 and f_2 . Since it rejects a band of frequencies, it is called a band-reject filter.

Application-wise, the filters discussed above have a very wide scope in the analogue domain. Let us focus on the design of filters in the digital domain, though functionality-wise the job of filters in both the domains remains

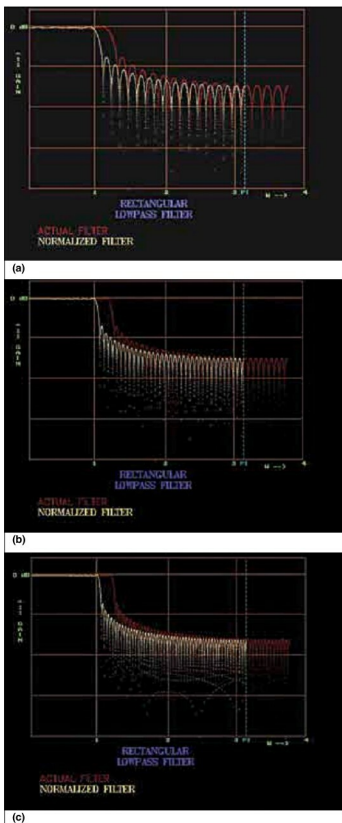


Fig. 2: Frequency response of the low-pass filter for (a) $N=49$, (b) $N=99$ and (c) $N=149$